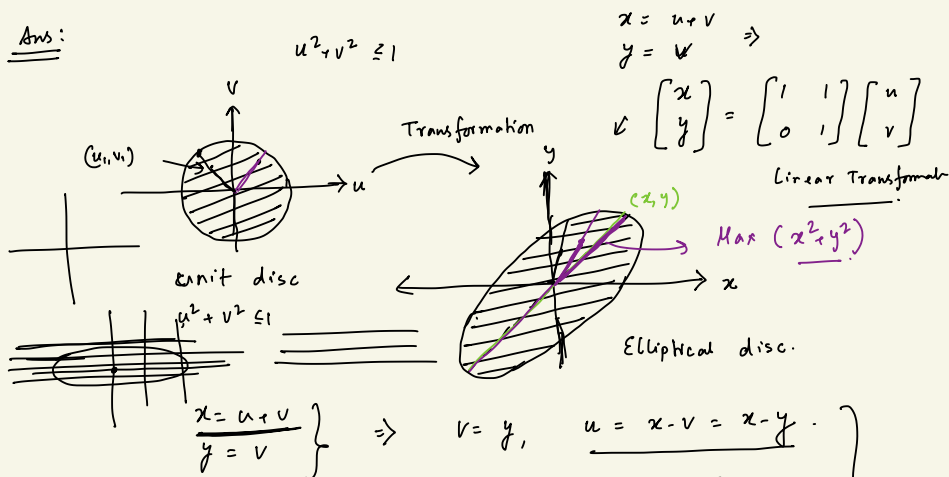


$$A = \{(x, y) : x = u + v, y = v, u^2 + v^2 \leq 1\}$$

Derive the length of the longest line segment that can be enclosed inside the region A.

Ans:



$$u^2 + v^2 \leq 1 \Rightarrow y^2 + (x - y)^2 \leq 1$$

$$\Rightarrow x^2 - 2xy + 2y^2 \leq 1$$

↑
Ellipse

$$x^2 - 2xy + 2y^2 = 1$$

$$\left\{ \begin{array}{l} \frac{\partial}{\partial x} (x^2 - 2xy + 2y^2) = 0 \\ \frac{\partial}{\partial y} (x^2 - 2xy + 2y^2) = 0 \end{array} \right\} \Rightarrow \begin{array}{l} x = 0 \\ y = 0 \end{array}$$

$$\text{Max}(x^2 + y^2) = \text{Max}(u + v)^2 + v^2$$

$$= \text{Max}(u^2 + 2uv + 2v^2)$$

subject to the condition : $u^2 + v^2 \leq 1$

$$\text{Max}(u^2 + 2uv + 2v^2) \quad \text{subject to} \quad u^2 + v^2 \leq 1$$

Maximum will occur at the boundary:

at $u^2 + v^2 = 1$

$$\left[\begin{array}{l} \text{Max} : (u^2 + 2uv + 2v^2) \\ \text{Subject to : } u^2 + v^2 = 1 \end{array} \right]$$

1st Method: $u = \cos \theta, \quad v = \sin \theta$

$$\begin{aligned} & \underline{u^2 + 2uv + 2v^2} \\ &= 1 + \sin 2\theta + \sin^2 \theta \\ &= 1 + \sin 2\theta + \frac{1 - \cos 2\theta}{2} \\ &= \frac{3}{2} + \sin 2\theta + \frac{1}{2} \cos 2\theta \\ &\leq \frac{3}{2} + \sqrt{1 + \frac{1}{4}} \\ &= \frac{3}{2} + \frac{\sqrt{5}}{2} = \frac{3+\sqrt{5}}{2} \end{aligned}$$

$$\begin{aligned} \text{Max}(x^2+y^2) &= \frac{3+\sqrt{5}}{2} \Rightarrow \text{Max } 2\sqrt{x^2+y^2} \\ &= 2\sqrt{\frac{3+\sqrt{5}}{2}} \\ &= \sqrt{6+2\sqrt{5}} \\ &= \sqrt{(\sqrt{5}+1)^2} = \underline{\underline{\sqrt{5}+1}} \end{aligned}$$

Matrix Method: (Use only in MCQs)

$$\left[\begin{array}{c} x^2+y^2=1, \text{ Max } (3x^2+4xy+2y^2) \end{array} \right]$$

$$\underline{\text{Max: } u^2 + 2uv + 2v^2}$$

$$u^2+v^2=1$$

$$[u \ v] \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$$

$$\lambda^2 - 5\lambda + 2 = 0$$

$$\frac{5 \pm \sqrt{17}}{2}$$

Calculate the eigenvalues

Max eigenvalue is the required maximum value

$$\lambda^2 - 3\lambda + 1 = 0$$

$$\Rightarrow \lambda = \frac{3 \pm \sqrt{5}}{2}$$

$$\text{Max } \frac{3+\sqrt{5}}{2}$$

Quadratic Form optimization

Q. Let, $x^2+y^2+z^2=1$. Find $Q(x,y,z)$

$$\text{Max } \left(\frac{x^2}{1} + \frac{y^2}{2} + \frac{z^2}{3} + 4xy + 6yz + 10zx \right)$$

$$\begin{matrix} x \\ y \\ z \end{matrix} \begin{bmatrix} 1 & 2 & 5 \\ 2 & 2 & 3 \\ 5 & 3 & 3 \end{bmatrix}$$

Find eigenvalues

Max eigenvalue

= max value of $Q(x,y,z)$

$\square$ Maximize $(u^2 + 2uv + 2v^2)$
 Subject to $(u^2 + v^2 = 1)$

$\text{Max } (x^2 + y^2) = k$
 Subject to
 $3x^2 + 4xy + 9y^2 = 1$

(x, y)
 Maximize : $k = x^2 + y^2$
 Subject to $x^2 - 2xy + 2y^2 = 1$

We shall find the maximum k s.t
 $x^2 + y^2 \leq k \cdot 1$ [I want this condition to hold for all values of x, y]

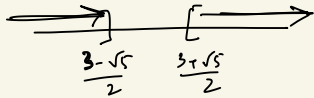
$$\Rightarrow x^2 + y^2 \leq k(x^2 - 2xy + 2y^2)$$

$$\Rightarrow (k-1)x^2 - 2kxy + (2k-1)y^2 \geq 0$$

$$\Rightarrow (k-1)\left(\frac{x}{y}\right)^2 - 2k\left(\frac{x}{y}\right) + (2k-1) \geq 0$$

$$\Rightarrow D \leq 0 \Rightarrow 4k^2 - 4(k-1)(2k-1) \leq 0$$

$$\Rightarrow k^2 - 3k + 1 \geq 0$$

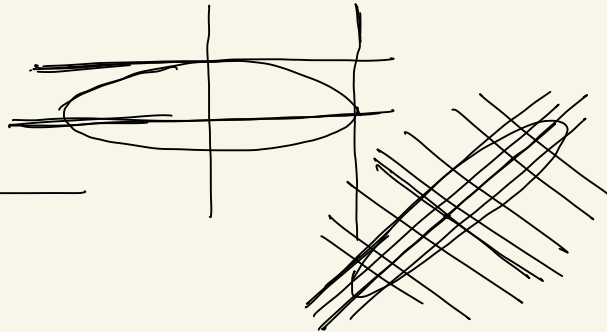


$$k \geq \frac{3+\sqrt{5}}{2}$$

$$k \leq \frac{3-\sqrt{5}}{2}$$

$\square$

$$\frac{dy}{dx} = 0$$



$$\frac{\partial S}{\partial x} = 0$$

Along which direction, the gradient (slope) is 0 ^{along the x-axis in one sense}

$$\frac{\partial S}{\partial y} = 0$$

direction $\parallel$ to the minor axis

direction $\parallel$ to the major axis of the ellipse

□

$$a_1 \geq a_2 \geq a_3 \geq \dots \geq a_n$$

$$f(a_1, \dots, a_n) \geq g(a_1, \dots, a_n)$$

Rearrangement inequality: $a_1 \leq a_2 \leq \dots \leq a_n$
 $b_1 \leq b_2 \leq \dots \leq b_n$

Chebyshev inequality (discrete)

$$\frac{\sum_{i=1}^n a_i b_i}{n} \geq \frac{\sum_{i=1}^n a_i}{n} \cdot \frac{\sum_{i=1}^n b_i}{n}$$

$$\Rightarrow \sum_{i=1}^n a_i b_i \geq \left(\frac{1}{n}\right) \sum_{i=1}^n a_i \sum_{i=1}^n b_i$$

Continuous form:

$[g, h: [a, b] \rightarrow \mathbb{R}]$ similarly sorted
 either both of them non-decreasing
 or both of them non-increasing

$$\int_a^b g(u) h(u) du \geq \left(\frac{1}{(b-a)}\right) \int_a^b g(u) du \int_a^b h(u) du$$

⑦

□

Let $f: [0, \infty) \rightarrow \mathbb{R}$ non-decreasing cont.

$$(z-x) \int_y^z f(u) du \geq (z-y) \int_x^z f(u) du$$

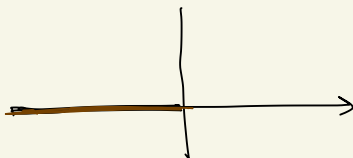
$$\Leftrightarrow \frac{1}{z-x} \cdot (z-y) \int_x^z f(u) du \leq \int_y^z f(u) du$$

$$= \frac{1}{z-x} \int_y^z 1 \cdot du \cdot \int_x^z f(u) du$$

Let,

$$h(u) = \begin{cases} 0, & \text{if } x \leq u < y \\ 1, & \text{if } y \leq u \leq z \end{cases}$$

$$\int_x^z f(u) du = \int_x^y 0 \cdot f(u) du + \int_y^z 1 \cdot f(u) du = \int_y^z f(u) du$$



h is non-decreasing
in (x, z)

f is also non-decreasing

$$\int_x^z \underbrace{f(u) h(u)}_z du \geq \frac{1}{z-x} \left(\int_x^z f(u) du \right) \left(\int_x^z h(u) du \right)$$

$$\Rightarrow \int_y^z f(u) du \geq \frac{1}{z-x} \int_x^z f(u) du \int_y^z 1 \cdot du$$

$$\Rightarrow \int_y^z f(u) du \geq \frac{z-y}{z-x} \int_x^z f(u) du$$

$$\Rightarrow (z-x) \int_y^z f(u) du \geq (z-y) \int_x^z f(u) du$$

$x^n, e^x \nearrow (0, 1)$

Chebyshev
Ineq:

$$\int_0^1 x^n \cdot e^x dx \geq \frac{1}{1-0} \int_0^1 x^n dx \cdot \int_0^1 e^x dx$$

$$= \frac{1}{n+1} \cdot (e-1) = \frac{e-1}{n+1}$$